Design and Simulation Quiz

- Next Week إن شاء الله
- Bring your Laptop
- Material included in quiz can be found in the Recordings Section of LMS page of your section.

Common Emitter Amplifier

EE 312

Experiment 10

Turki Almadhi, Dept. of EE, King Saud University

P-N Junction Cutoff State (Revision)

- A PN Junction is Said to be Cutoff When the Potential of the P-side relative to the N-side is Less than the Cut-In (Turn-On) Voltage
- Note that a Slightly Forward P-N Junction is Considered Cutoff as Long as the Forward Voltage Stays Less than the Cut-In Voltage.

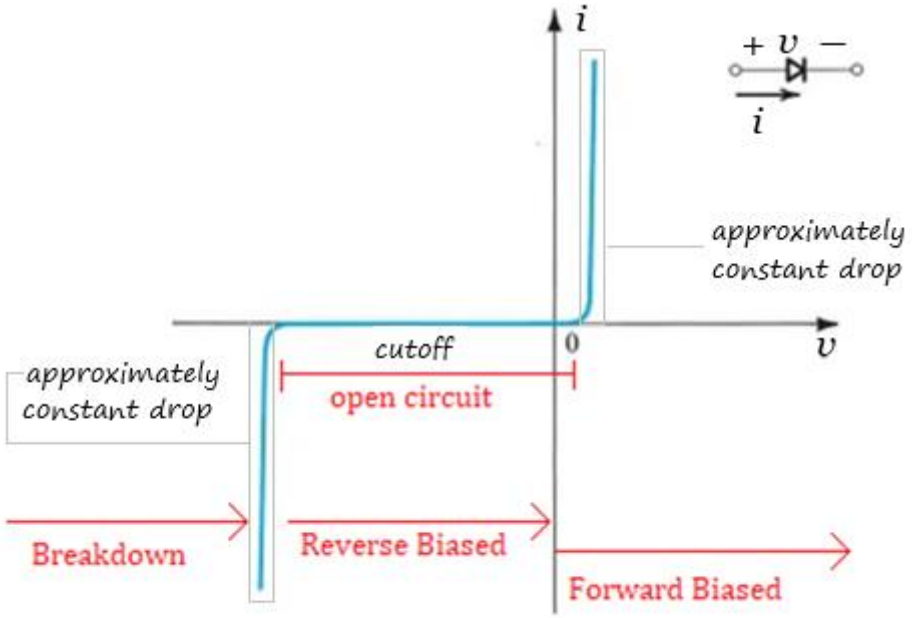


Figure 4.7 The i–v characteristic of a silicon junction diode.

Summary of Modes of Operation for the npn BJT (Revision)

MODE	EBJ Bias Condition	CBJ Bias Condition	Circuit Model	Application
Cutoff	VBE < VBE(on) Cutoff	VBC < VBC(on) Cutoff	*Open switch* between the collector and the emitter	Switching Circuits, like logic gates
Saturation	VBE ≥ VBE(on) Forward Biased	VBC ≥ VBC(on) Forward Biased	*Closed switch* (with offset of about 0.2 V) between the collector and the emitter	Switching Circuits, like logic gates
Active	VBE ≥ VBE(on) Forward Biased	VBC < VBC(on) Cutoff	***A PN junction*** between the base and the emitter and a ***Voltage-Controlled Current Source*** between the collector and the emitter	Amplifier
VBE(on) ≈ 0.5 V and VBC(on) ≈ 0.4 V				

Revision of Modes of Operation

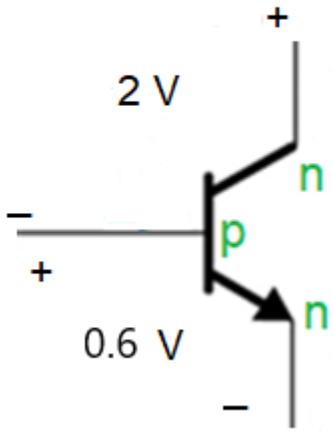
Based on the *bias condition of its two junctions*, three distinct modes of operation are obtained:

Switching Applications (e.g., logic circuits).

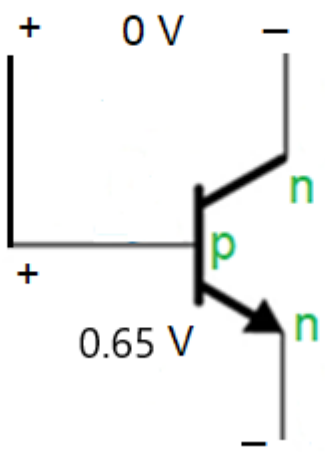
- The **cutoff** mode (both junctions cutoff) [open switch]
- The **saturation** mode (both junctions turned on) [closed switch]

- The **active mode** (EBJ turned on and the CBJ is cutoff) is employed in amplifier circuits.

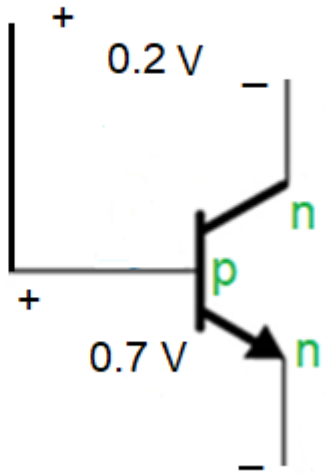
In Which Mode are Theses Transistors?



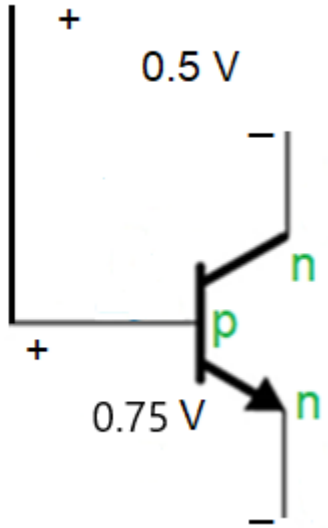
Active



Active



Active



Saturation

Objectives

- To assemble a common emitter amplifier circuit, perform dc and small signal measurements, and calculate some important parameters.

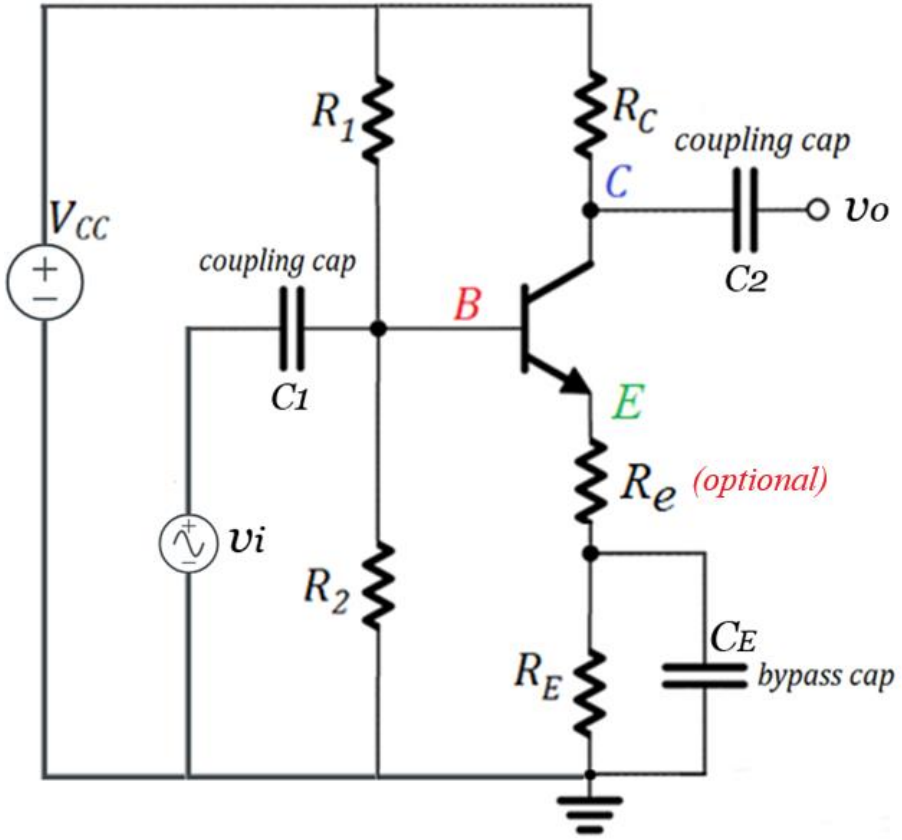
- Examine distortion caused by an input signal having too large amplitude.

- Learn how to size the four resistors to have a stable Q point in the *active mode* despite potential variations in temperature and/or β.

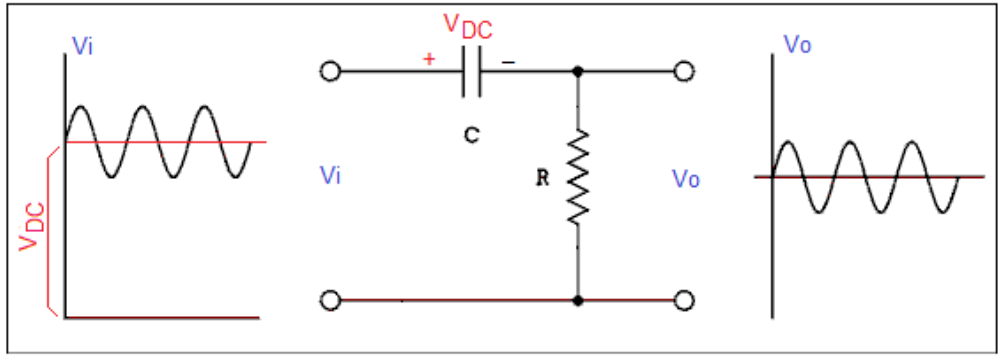
The Common Emitter Amplifier

The CE Amplifier

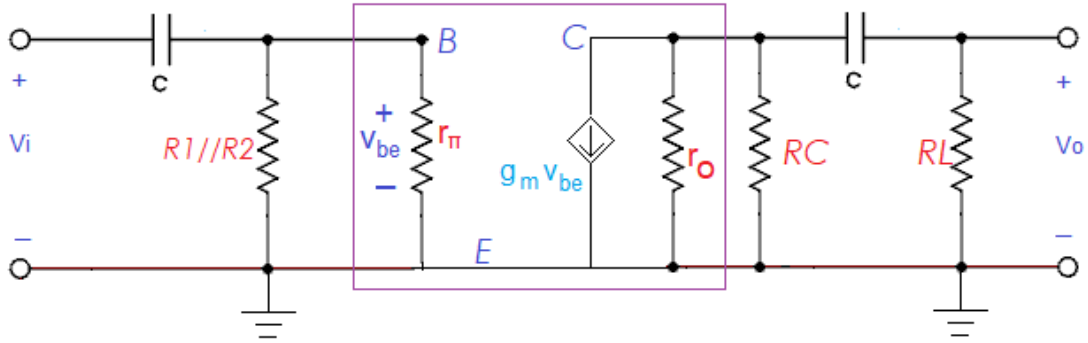
- The common emitter (CE) amplifier is the most famous of the three BJT amplifier configurations.
- The discrete CE amplifier receives the input signal at the base (via a coupling capacitor) and delivers the output from the collector.
- The other two being the common base (used to achieve higher bandwidth) and the common collector (used as a voltage buffer).



Coupling (DC Blocking) Capacitor Acts as a HP Filter



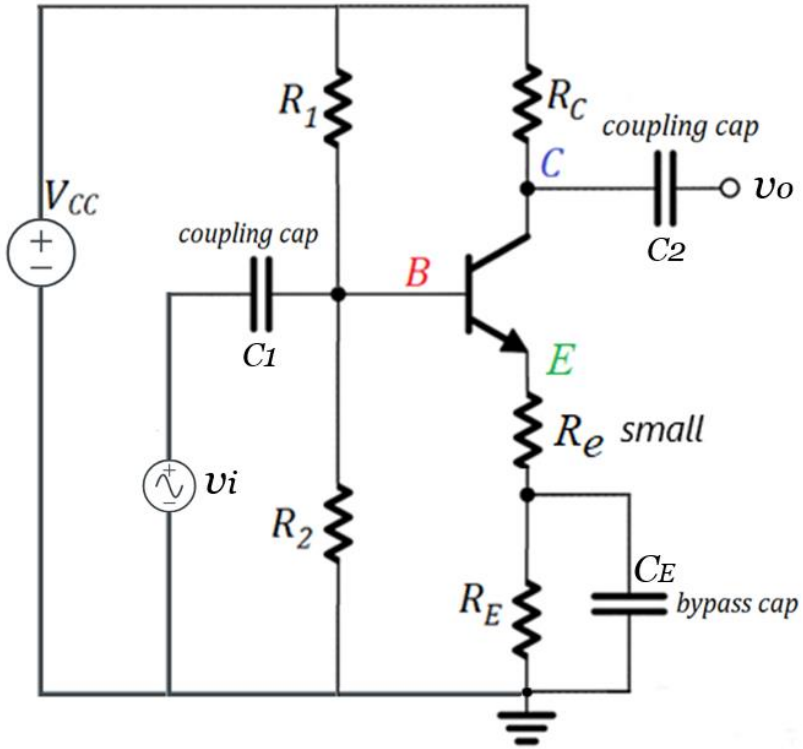
HIGH PASS FILTER



The Small Signal Equivalent Circuit for the CE Amplifier

Coupling and Bypass Capacitors

- **Coupling** capacitors (**also called dc blocking**), are used to act as short circuit at signal frequencies while blocking dc.
- **Bypass** capacitor, C_E, is used in order to bypass (short out) R_E at signal frequencies; therefore, nearly eliminate its undesirable effect on the voltage gain A_v.



Roles of Various Components in the CE Amp

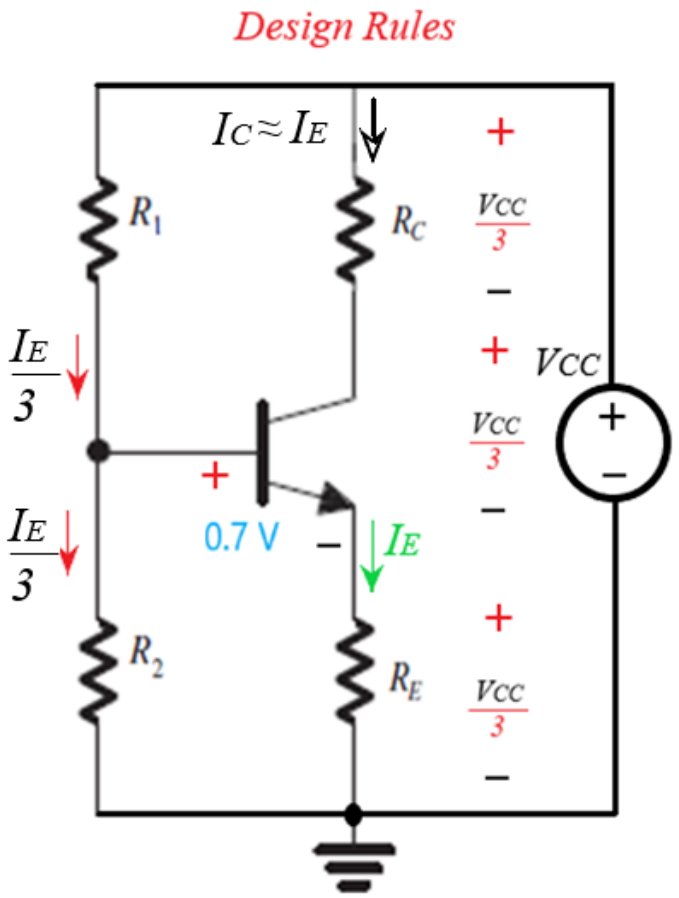
Component	Role
R_1 *and* R_2	Used to obtain appropriate dc base voltage V_B.
R_C	Essential to obtain voltage gain
R_E	Helps to *stabilize the operating point* through negative feedback. However, to prevent voltage gain loss ⇒ Use bypass capacitor in parallel with it.
R_e *(optional)*	Increases the amplifier's ***input resistance*** and extends its ***linear range***.
C_1	Couple the signal to the amplifier while blocking possible dc component in the signal
C_2	Removes any dc from the collector voltage while allowing the signal $v_c(t)$ to pass.
C_E	Bypasses R_E at signal frequencies

Sizing R_C, R_E, R1 and R2

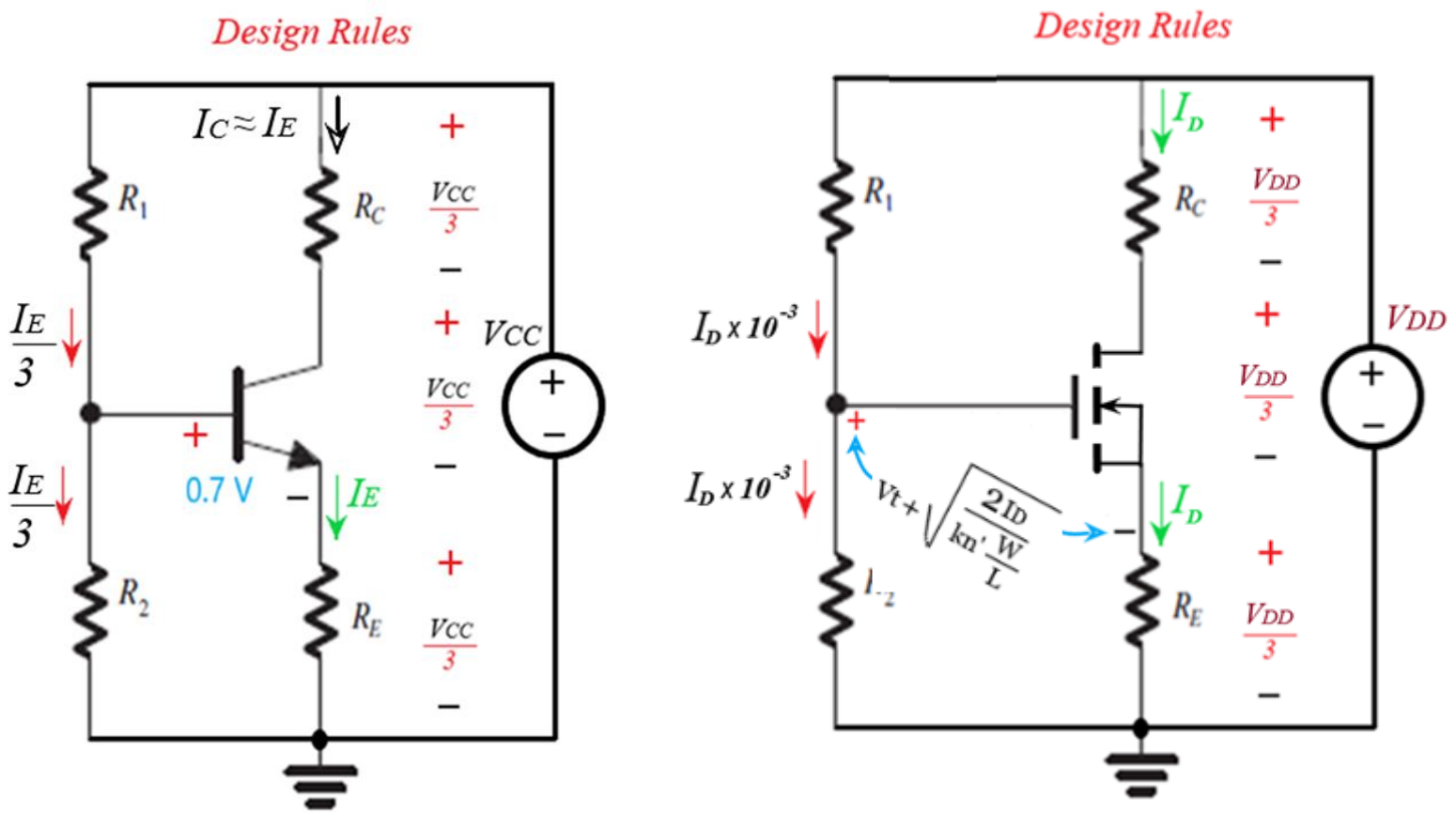
To obtain a Stable Operating Point Q (I_{CQ}, V_{CEQ}) Fairly Independent of β and Temperature:

Given: $V_{BE(on)} \approx 0.7\ V$ and a targeted I_E.

- *Start off with $V_{RC} = V_{CE} = V_{RE} =$ [V_{CC} / 3].*
- Given I_E, Calculate R_E from Ohm's *law*
- *Select $R_C = R_E$*
- *Calculate $V_B = V_{BE(on)} + V_E$ and $I_{R1} = I_{R2} =$ [I_E / 3].*
- *Calculate R1 and R2 from Ohm's law*
- *Use Appendix F to find the nearest standard resistors to those you have calculated.*



Comparing DC Biasing Design Rules
CE Amp vs. CS Amp



Example

- **Problem:** Design a 4-resistor bias circuit with given parameters.
- **Given data:** $I_E = 0.75\ mA$, $\beta = 100$, $V_{CC} = 15$ V, $V_{BE} = 0.7$ V
- **Unknowns:** V_B, voltages across R_E and R_C; values for R_1, R_2, R_C and R_E.
- **Analysis:** A common approach is to divide V_{CC} equally between V_{CE}, R_E and R_C. Thus, $V_E = 5$ V and $V_C = 10$ V.

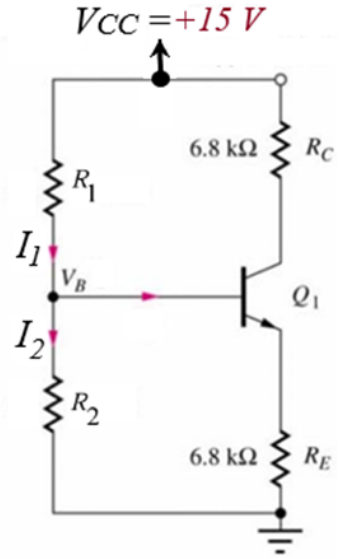
$$R_C \approx R_E = \frac{V_E}{I_E} = 6.667\,\text{k}\Omega \quad \Rightarrow \textit{Nearest Standard} = 6.65\,\text{k}\Omega$$

$$V_B = V_E + V_{BE} = 5.7\,\text{V}$$

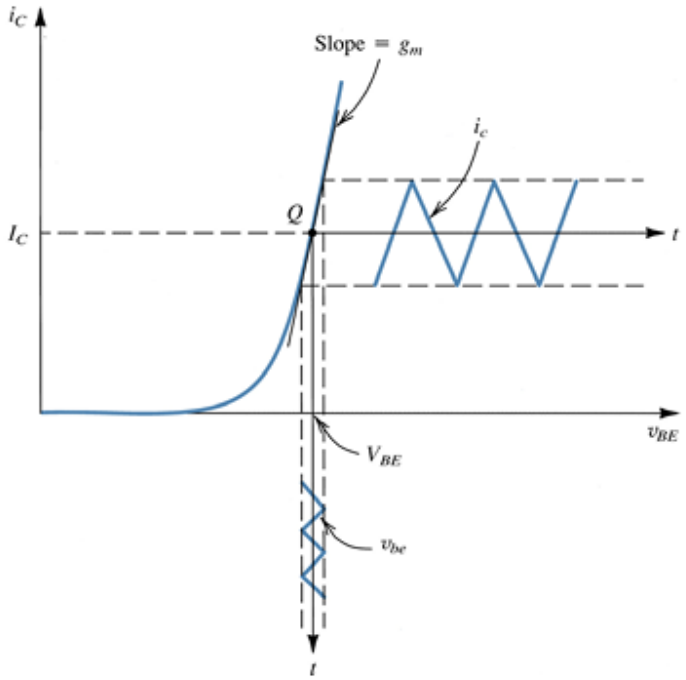
$$R_2 = \frac{V_B}{(I_E/3)} = 37.2\ \text{k}\Omega \quad \Rightarrow \textit{Nearest Standard} = 37.4\,\text{k}\Omega$$

$$R_1 = \frac{V_{CC} - V_B}{(I_E/3)} = 22.8\ \text{k}\Omega \quad \Rightarrow \textit{Nearest Standard} = 22.6\,\text{k}\Omega$$

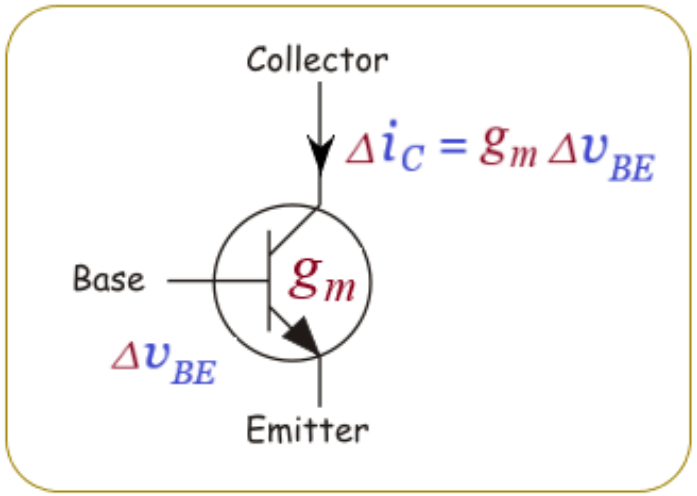
$0.1I_E \longleftrightarrow 0.33\,I_E$



Small Signal Variation is Necessary For Linear Operation



$$g_m \equiv \left.\frac{i_c}{v_{be}}\right|_{v_{be}\to 0} = \left.\frac{\partial i_C}{\partial v_{BE}}\right|_{i_C = I_C}$$



Factors Affecting the Voltage Gain Gv

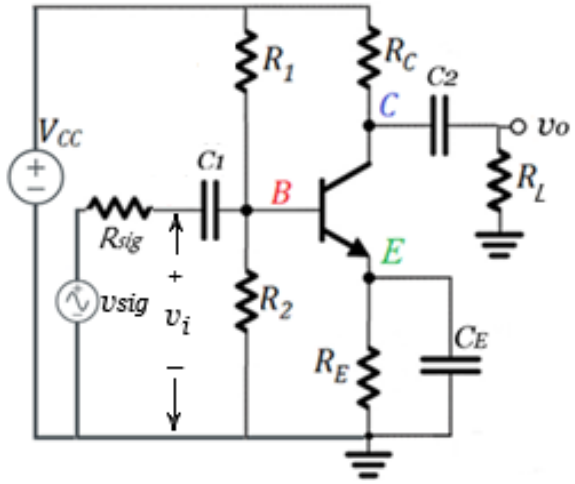
The Voltage Gain for CE Amplifier :

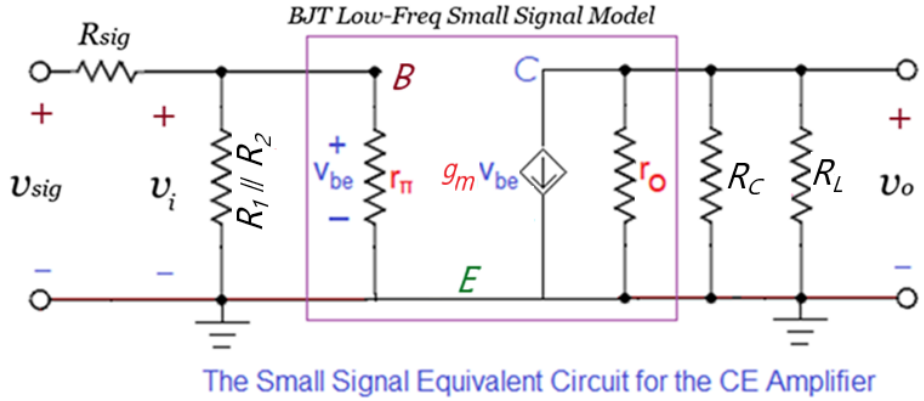
Assuming $R_{sig} << (R_1 \parallel R_2 \parallel r_\pi)$

$$G_v \equiv \frac{v_o}{v_{sig}} \cong A_v \equiv \frac{v_o}{v_i} = -g_m (R_C \parallel R_L \parallel r_o)$$
$$\cong -g_m (R_C \parallel R_L) \quad if\ (R_C \parallel R_L) << r_o$$
$$\cong -g_m R_C \quad if\ R_C << (r_o \parallel R_L)$$

$$g_m = \frac{I_C}{V_T}; \qquad [V_T \approx 26\ mV\ at\ T = 300\ K]$$

g_m and r_o are small signal parameters of the BJT, but they are strongly dependent on the DC Operating point, i.e., dc collector currrent I_C!

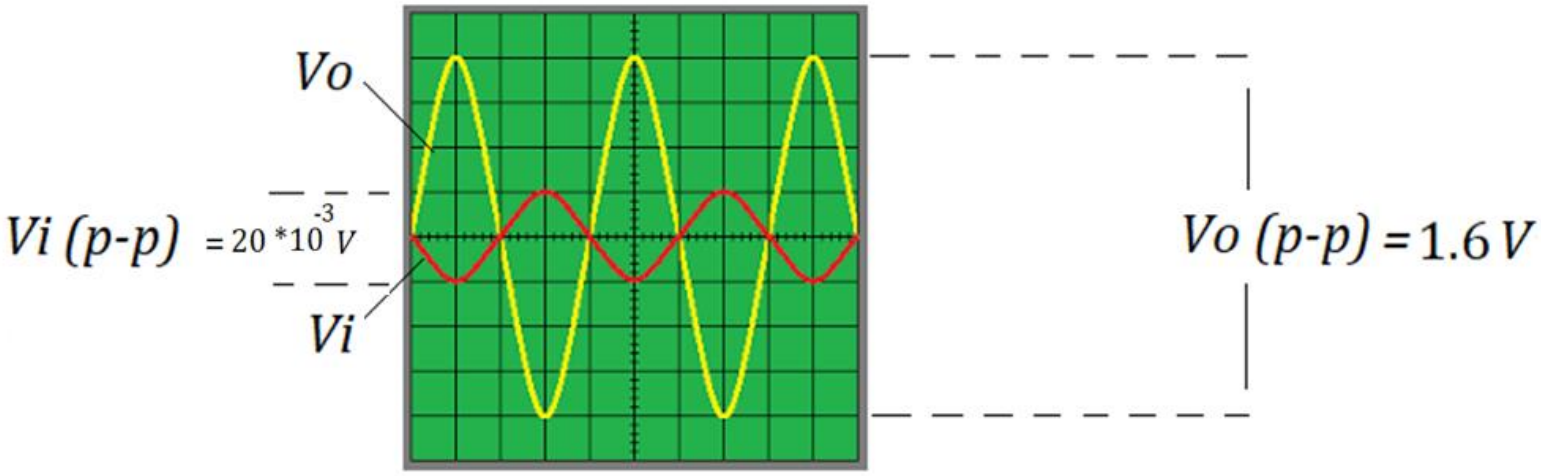




The Small Signal Equivalent Circuit for the CE Amplifier

How to Find the voltage gain *Av*

Example



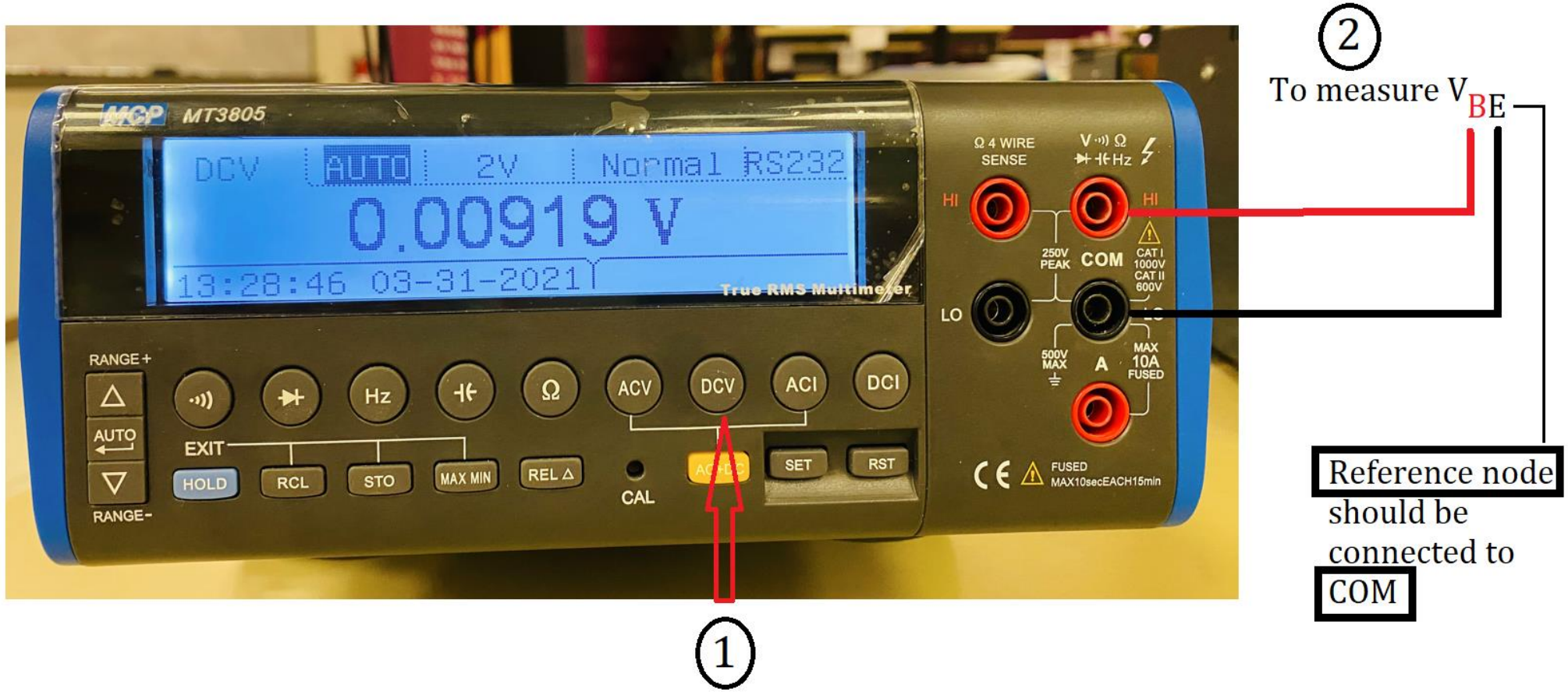
$$Av = \frac{Vo\ (p\text{-}p)}{Vi\ (p\text{-}p)} = \frac{1.6}{20 * 10^{-3}} = 80\ \text{V/V}$$

The Experimental Part

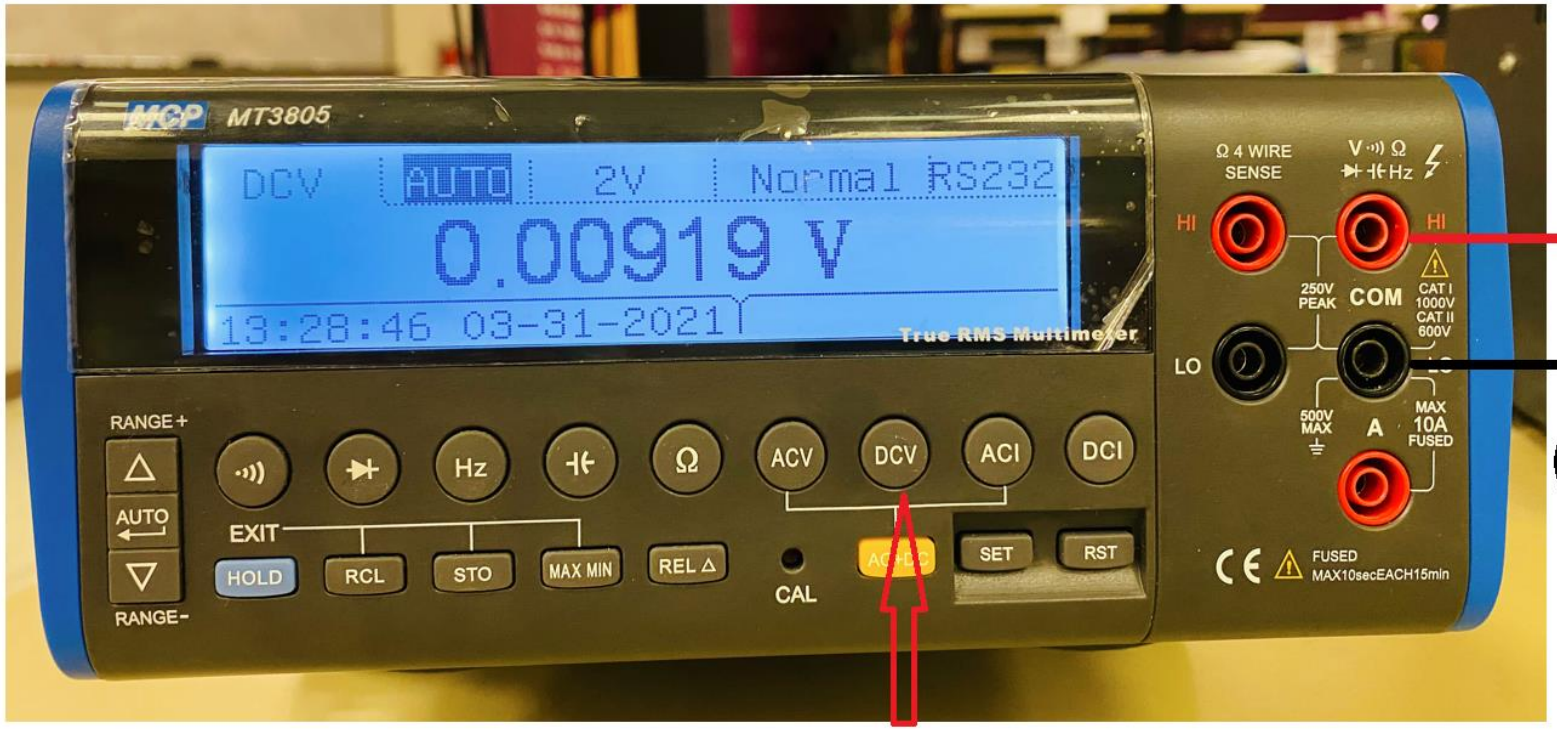
Measuring The *Actual* Resistances

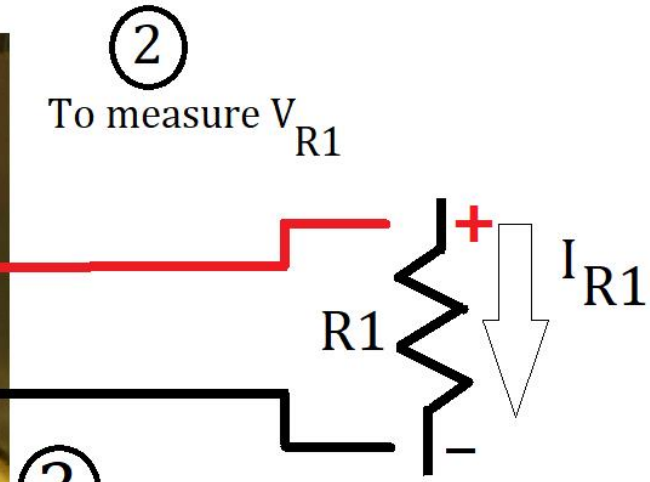


How to Measure DC Voltage



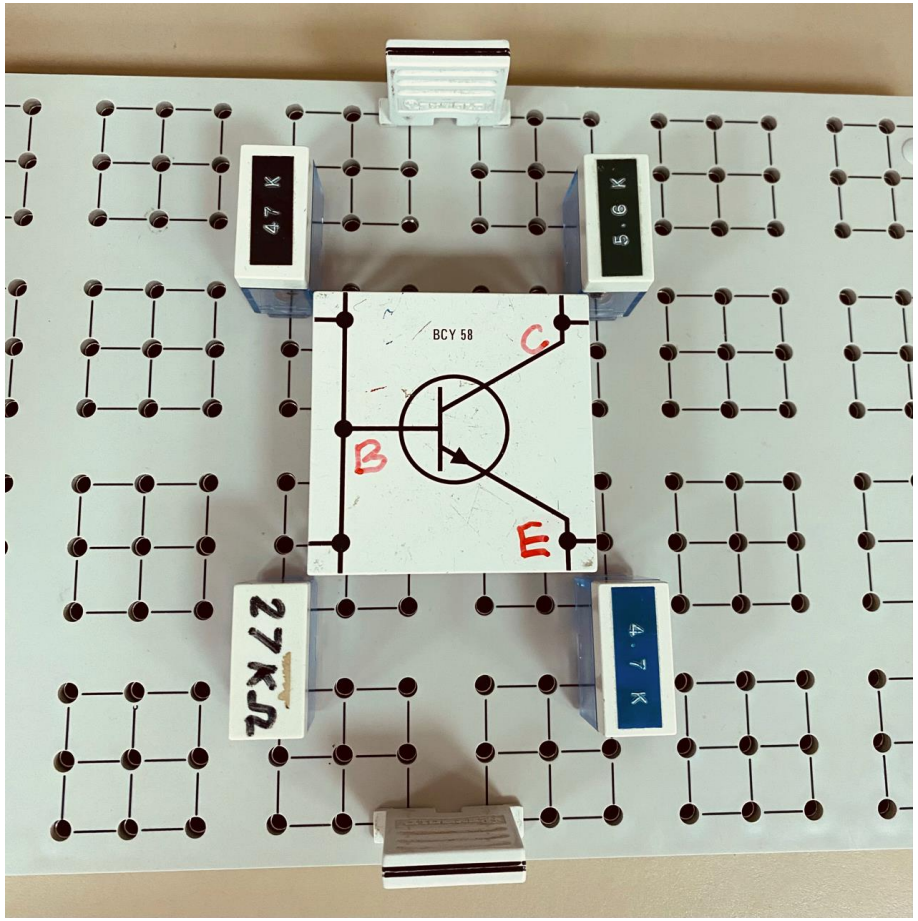
Using Ohm's Law to Obtain a Resistor's Current From Its Measured Voltage Drop

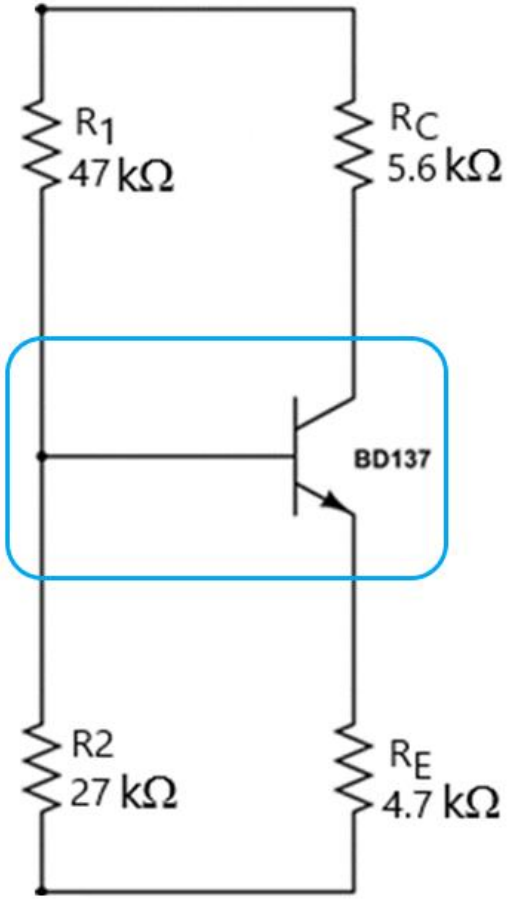




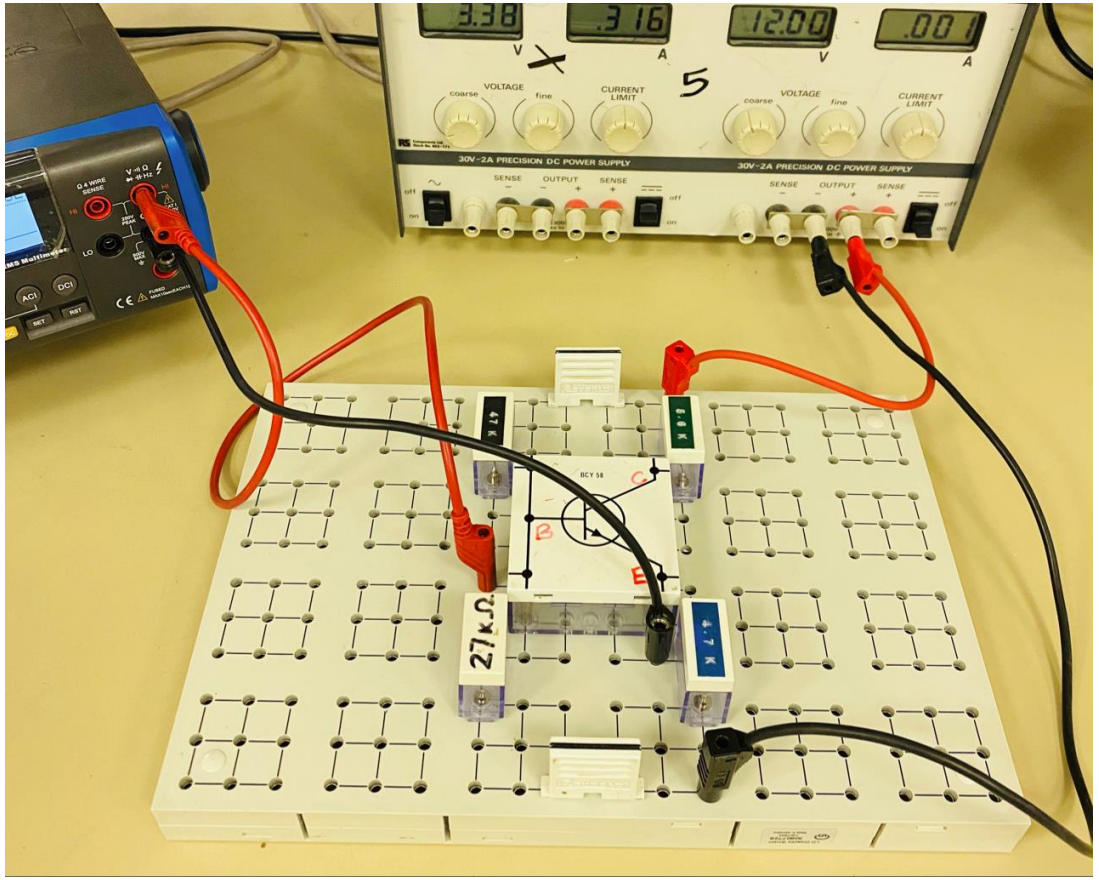
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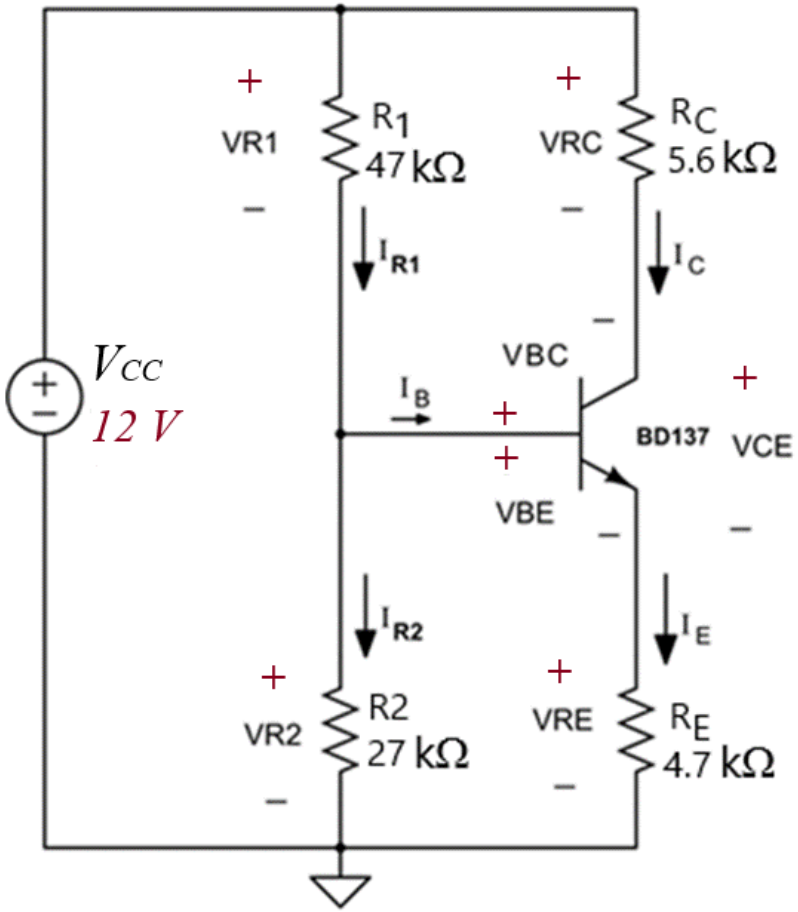
The Biasing Circuit without the dc Supply



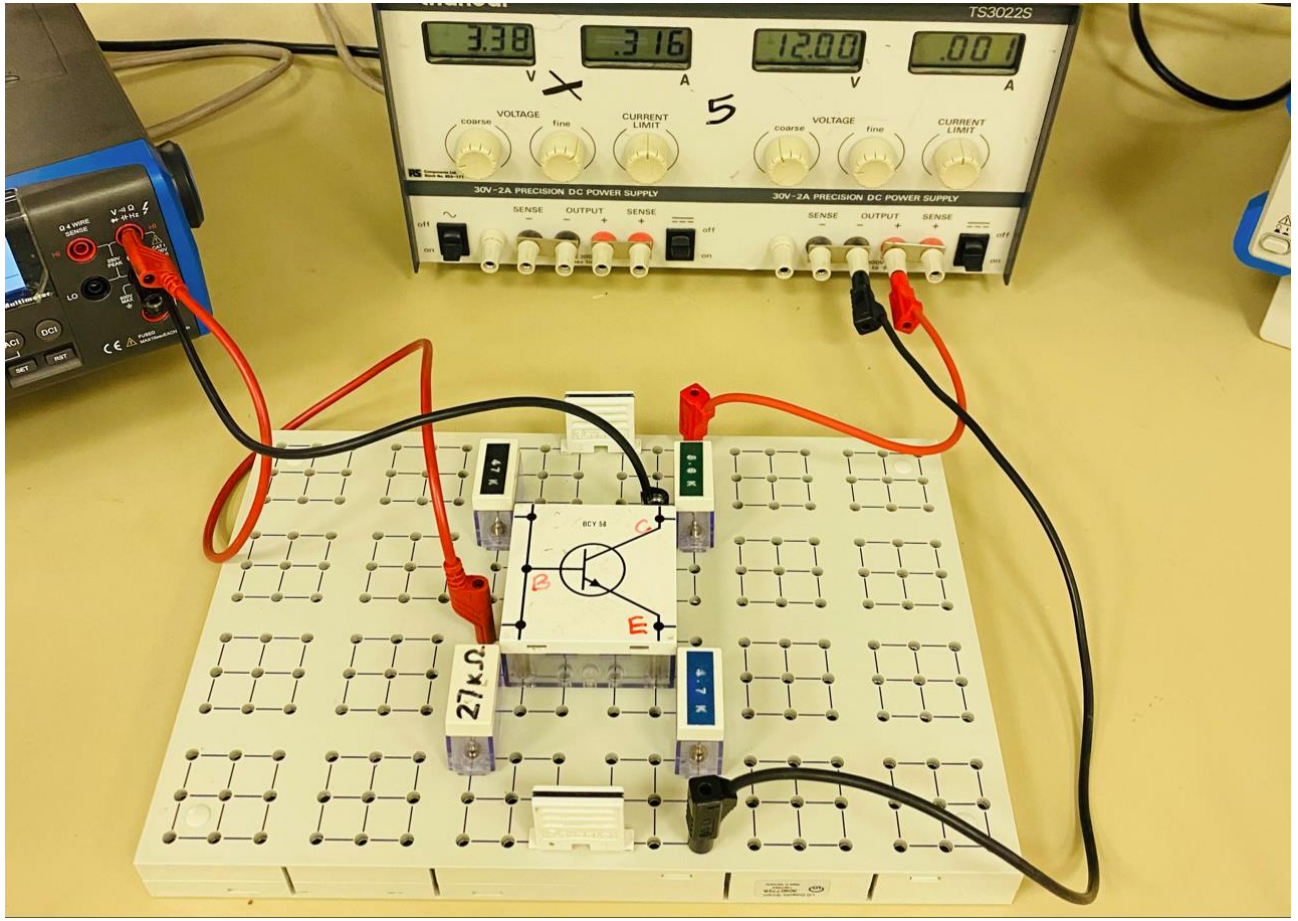


Adding V_{CC} then Measuring V_{BE}





Measuring V_{BC}



How to calculate Currents and Verify Voltages

$$I_C = \frac{V_{RC}}{R_C} \qquad I_E = \frac{V_{RE}}{R_E}$$

$$I_{R1} = \frac{V_{R1}}{R_1} \qquad I_{R2} = \frac{V_{R2}}{R_2}$$

$$I_B = I_{R1} - I_{R2} \qquad \beta = \frac{I_C}{I_B}$$

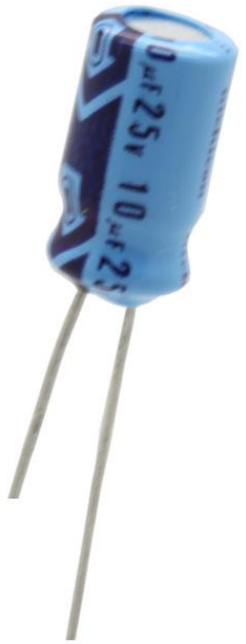
$$V_{RC} + V_{CE} + V_{RE} = V_{CC}$$

$$V_{R1} + V_{R2} = V_{CC}$$

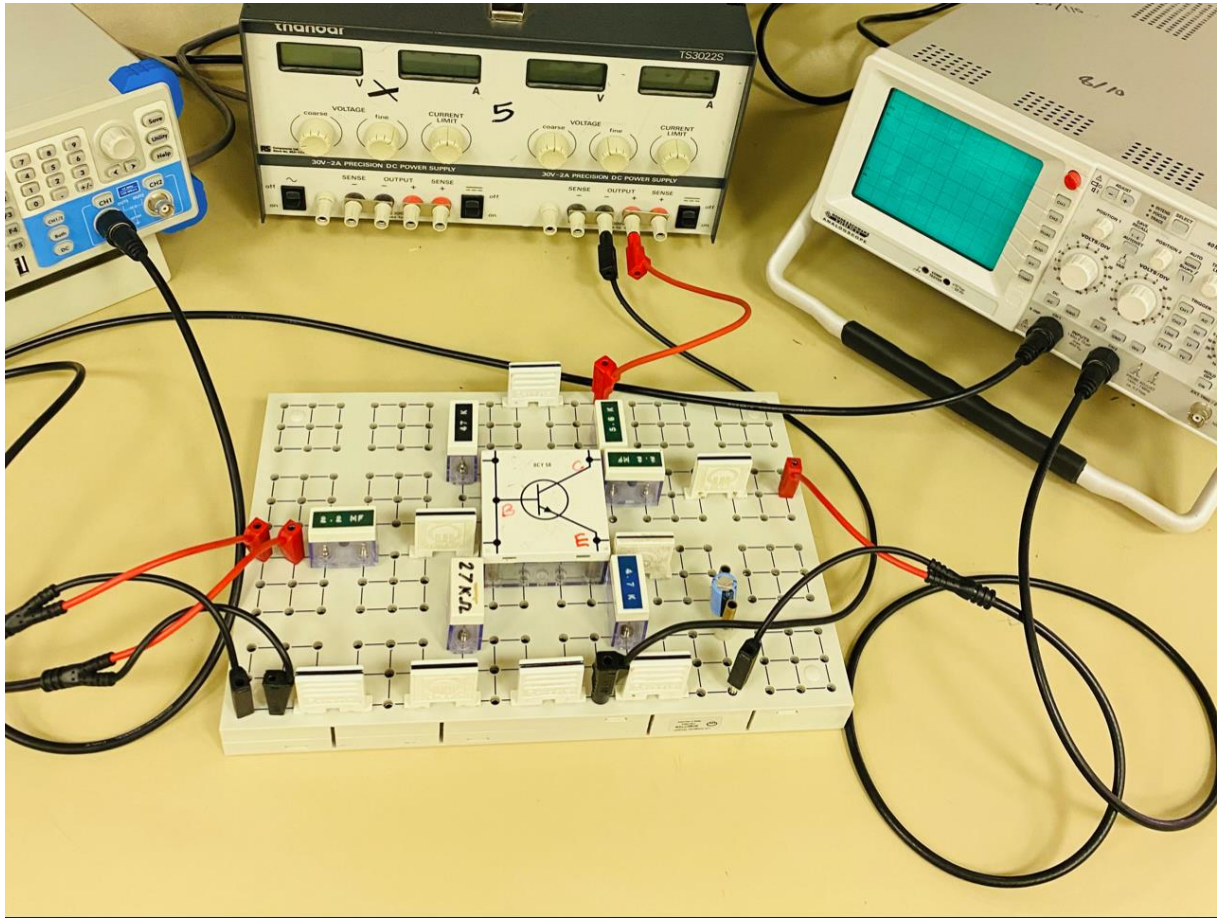
$$V_{CE} = V_{BE} - V_{BC}$$

Electrolytic Capacitor Polarity





The Complete Amplifier Setup



دعواتي لكم بالتوفيق

مع تقديري لجهودكم وتعاونكم

References

- A. S. Sedra and K. C.Smith, *Microelectronic Circuits*, New York: Oxford University Press, 2010.
- T. A. Almadhi, EE 312 Lab Manual:

https://lms.ksu.edu.sa/bbcswebdav/users/talmadhi/EE312/EE312.pdf